

(IX) Program Code

Consider the string: 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7 with frame size 3, calculate page fault using LRU algorithm.

Note: Attached the code at the end.

This practical can be performed in any of the compiler like Turbo C/ vi editor etc or Python Interpreter / IDE like any open source IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(X) Conclusion

Successfully implemented C/Python program on Least Recently Used (LRU) Page Replacement algorithm.

(XI) Practical Related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

- (1) Compare FIFO and LRU.
- (2) State the advantages and disadvantages of LRU.
- (3) Count number of page hits and page faults in above mention example given in section IX.
- (4) You are given the following page reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 Using 3 page frames, how many page faults occur with LRU?

[Space to Write for Answers]

1)	First In First Out (FIFO)	Least Recently Used (LRU)
1.	Replaces the oldest page in memory.	Replaces the least recently used page.
2.	Does not consider how recently a page was used.	Considers recent page usage before replacement.
3.	Simple & easy to implement.	More complex to implement.
4.	Lower implementation overhead.	Usually results in fewer page faults and better performance.

2) Advantages :-

1. Reduces page faults compared to FIFO in most cases.
2. Considers recent page usage, so frequently used pages are less likely to be replaced.
3. Provides better overall performance than FIFO for many workloads.

Disadvantages :-

1. More complex to implement than FIFO.
2. Requires tracking page usage, which increases overhead.
3. Higher implementation cost due to maintaining usage history.

3) Given :-

Reference String :- 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7

Frame Size :- 3

Page Replacement algorithm :- LRU

RS	0	1	2	3	0	1	2	3	0	1	2	3	4	5	6	7
F ₁	0	0	0	3	3	3	2	2	2	1	1	1	4	4	4	7
F ₂		1	1	1	0	0	0	3	3	3	2	2	2	5	5	5
F ₃			2	2	2	1	1	1	0	0	0	3	3	3	6	6
H/S	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Page Hits = 0

Page Faults = 16

4) Given :-

Reference String :- 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Page Frames :- 3

Page Replacement algorithm :- LRU

RS	1	2	3	4	1	2	5	1	2	3	4	5
F ₁	1	1	1	4	4	4	5	5	5	3	3	3
F ₂		2	2	2	1	1	1	1	1	1	4	4
F ₃			3	3	3	2	2	2	2	2	2	5
H/F	F	F	F	F	F	F	F	H	H	F	F	F

Page Hits = 2

Page Faults = 10